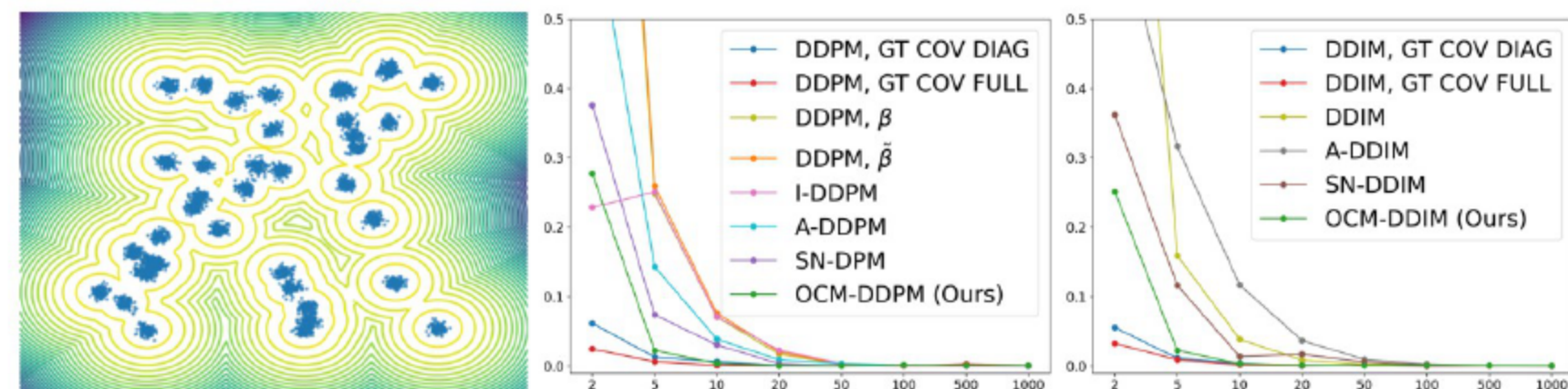


# Improving Probabilistic Diffusion Models with Optimal Diagonal Covariance Matching

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## 01 Problem: fixed variance slows diffusion



- Fixed or heuristic variances force very large step counts to reach strong sample quality and likelihood
- **Learning state-dependent covariance** cuts function evaluations while maintaining fidelity
- Our objective: estimate a better covariance to accelerate sampling without worsening negative log likelihood
- Focus on diagonal covariance for scalability to high-dimensional images

[Sohl-Dickstein et al., 2015; Ho et al., 2020; Song & Ermon, 2019]  
[Nichol & Dhariwal, 2021; Bao et al., 2022a]

## 02 DDPM/DDIM and why covariance matters

Modeling Capability

| Covariance Type                                                | + #Passes | Intuition                     |
|----------------------------------------------------------------|-----------|-------------------------------|
| $x_t$ -independent Isotropic $\beta$ (Ho et al., 2020)         | 0         | Cov. of $q(x_t x_{t-1})$      |
| $x_t$ -independent Isotropic $\tilde{\beta}$ (Ho et al., 2020) | 0         | Cov. of $q(x_t x_{t-1}, x_0)$ |
| $x_t$ -independent Isotropic Estimation (Bao et al., 2022b)    | 0         | Estimate from data            |
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| $x_t$ -dependent Diagonal OCM (Ours)                           | 1         | Learn from score              |
| $x_t$ -dependent Diagonal Estimation (Zhang et al., 2024b)     | $2M$      | Estimate from score           |
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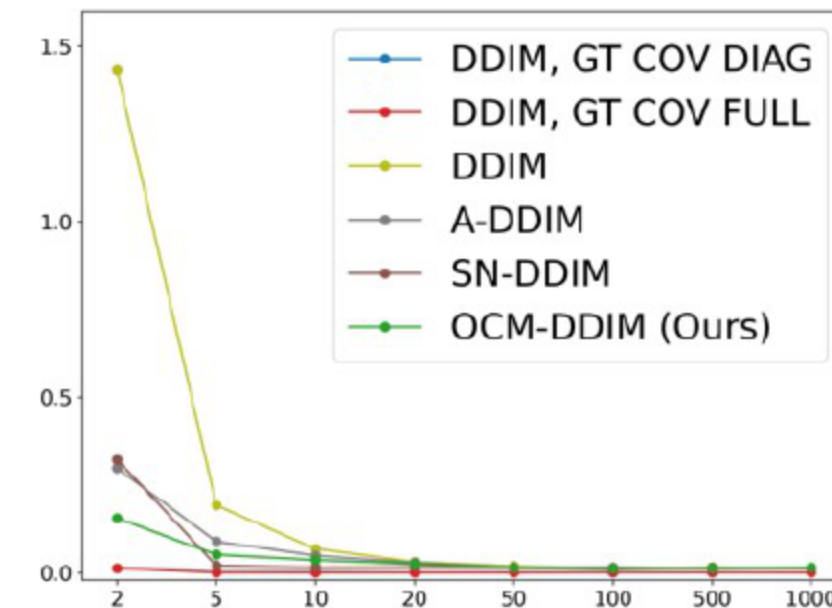
- When the step budget is small, the chosen covariance of the denoising distribution strongly impacts outcomes
- Heuristics such as using forward or posterior variances behave similarly only with very large steps
- **Goal:** improve covariance for fewer evaluations in DDPM and DDIM, including the zero-step target in DDIM

[Nichol & Dhariwal, 2021; Song et al., 2020]

- **Poor covariance choices** degrade likelihood and diversity under aggressive skipping

## 03 Optimal covariance from the score

- A generalized covariance identity expresses the optimal posterior covariance as a functional of the score and its Hessian
- With scaled Gaussian convolution, the optimal state-dependent covariance has a closed form from the score
- **Practical choice:** use only the diagonal of the Hessian to gain **speedups** with strong accuracy
- Links to higher-order Tweedie ideas and optimality under divergence criteria

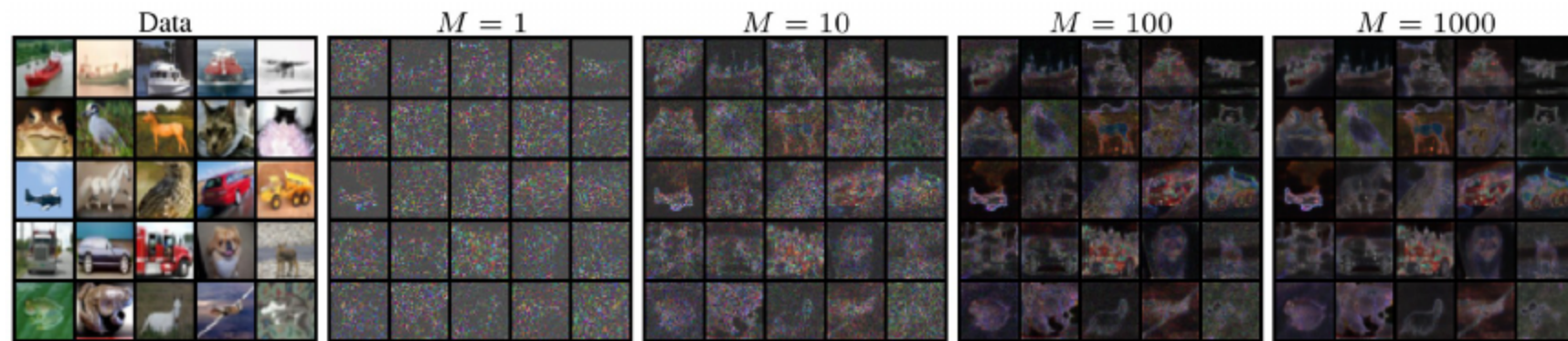


(d) DDIM MMD v.s. Steps

[Efron, 2011; Robbins, 1992; Zhang et al., 2024b]  
[Bao et al., 2022b]

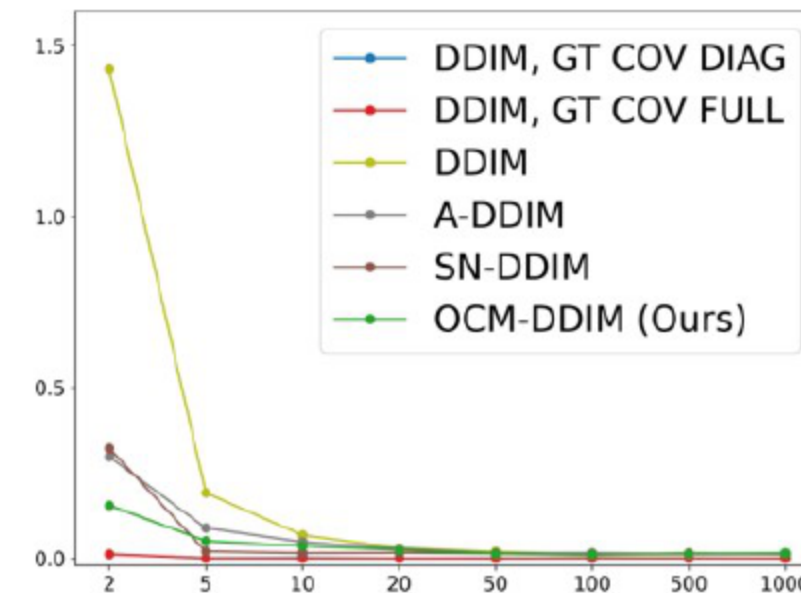


## 04 Hessian cost and diagonal estimation



- Exact Hessians demand quadratic memory and many network evaluations per denoising step
- Hutchinson or Rademacher estimators approximate the diagonal but need many probe vectors for low variance
- Large probe counts** imply many extra forward and backward passes, negating acceleration [He et al., 2020; Wang et al., 2020; Martens et al., 2012] [Zhang et al., 2024b]
- We need a way to predict the diagonal Hessian with **single-pass** overhead at inference

## 05 Unbiased Optimal Covariance Matching (OCM)



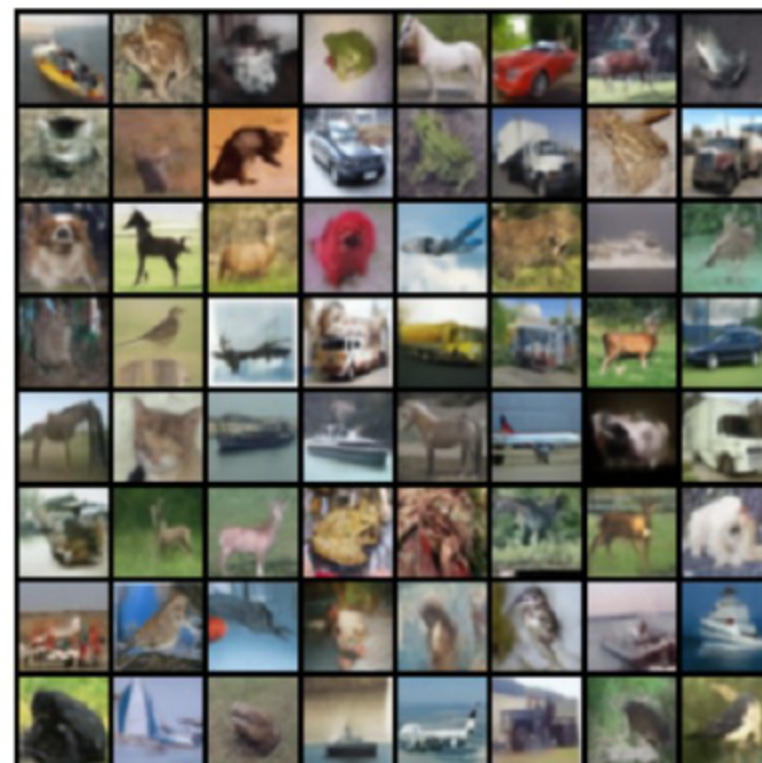
(d) DDIM MMD v.s. Steps

- Train a network to predict the diagonal of the score Hessian using an **unbiased** upper-bounding objective
- Monte Carlo over Rademacher vectors keeps the estimator unbiased and works well even with a single sample
- At inference, form a diagonal covariance from the predicted Hessian diagonal with **negligible** extra cost
- Reduces covariance error relative to data-only learning objectives

[Kingma & Welling, 2013; Dayan et al., 1995; Bao et al., 2022a] [Nichol & Dhariwal, 2021]

## 06 Applying OCM to skip-step DDPM/DDIM

- Learn the Hessian-diagonal head using a pretrained score network, then set diagonal covariance at each step
- Skip-step DDPM: parameterize a Gaussian transition with the usual mean and the learned diagonal covariance
- Skip-step DDIM: sample a non-degenerate image estimate with learned covariance, then update the latent state
- Works for both pixel and latent diffusion models



[Song et al., 2020; Vahdat et al., 2021; Rombach et al., 2022]

## 07 Shared-backbone Hessian head

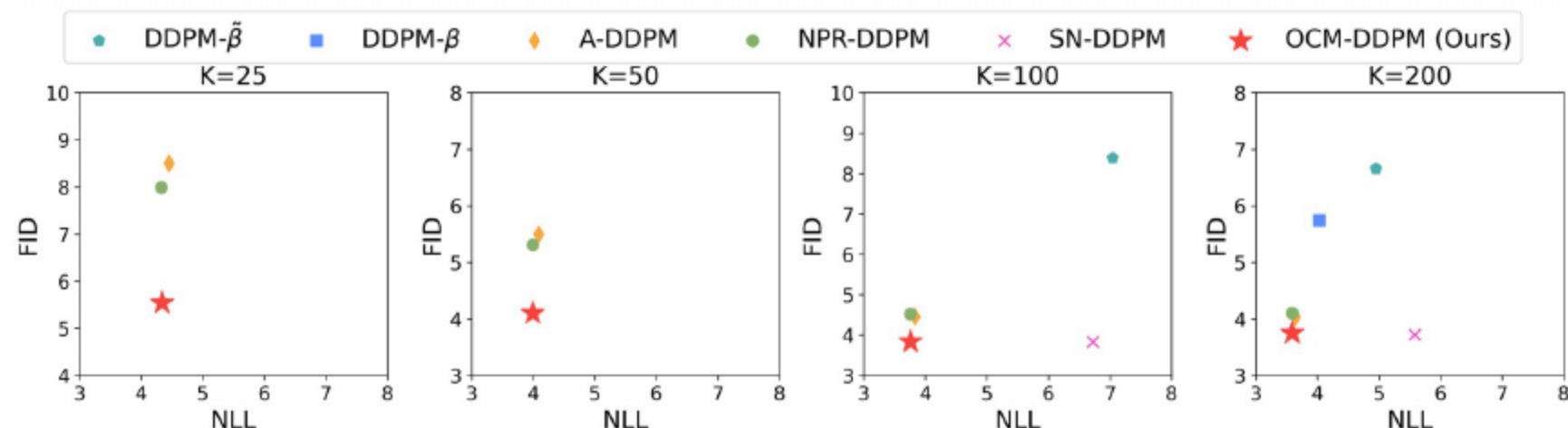
|                  | SCORE PREDICTION NETWORK | SCORE & SN PREDICTION NETWORKS | SCORE & DIAGONAL HESSIAN PREDICTION NETWORKS |
|------------------|--------------------------|--------------------------------|----------------------------------------------|
| CIFAR10 (LS)     | 52.54 M / 44.37 MS       | 52.55 M / 44.61 MS (+0.5%)     | 52.55 M / 44.51 MS (+0.3%)                   |
| CIFAR10 (CS)     | 52.54 M / 45.13 MS       | 52.55 M / 45.24 MS (+0.2%)     | 52.55 M / 45.23 MS (+0.2%)                   |
| CELEBA 64X64     | 78.70 M / 67.88 MS       | 78.71 M / 69.15 MS (+1.9%)     | 78.71 M / 68.89 MS (+1.5%)                   |
| IMAGENET 64X64   | 121.06 M / 106.58 MS     | 121.49 M / 112.53 MS (+5.6%)   | 121.49 M / 112.23 MS (+5.3%)                 |
| LSUN BEDROOM     | 113.67 M / 692.58 MS     | 114.04 M / 771.55 MS (+11.4%)  | 114.04 M / 770.73 MS (+11.2%)                |
| IMAGENET 256X256 | 675.13 M / 22.84 MS      | MISSING MODEL                  | 677.80 M / 23.01 MS (+0.7%)                  |

- Attach a lightweight Hessian-diagonal head to shared BaseNet features such as UNet or DiT
- Freeze the pretrained score network and train only the Hessian head with the OCM objective
- Minimal** compute and memory overhead; prediction runs in parallel with mean or score inference
- Simple to integrate into existing diffusion training pipelines

[Ho et al., 2020; Peebles & Xie, 2023]

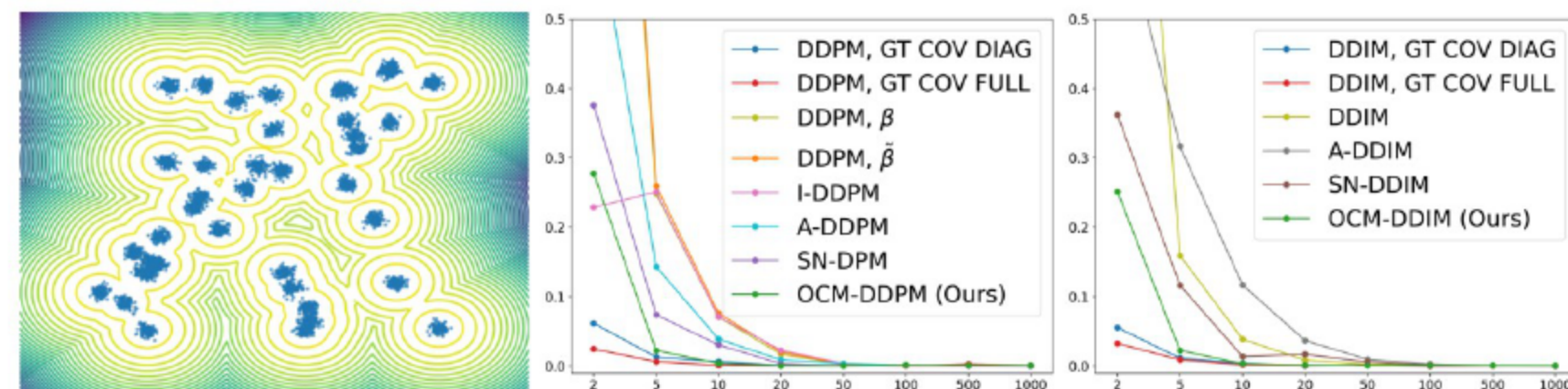


## 08 Preview: best FID-NLL trade-off



- On CIFAR-10 with the cosine schedule, OCM achieves the best frontier of sample quality versus likelihood
- Better likelihood than heuristic and learned baselines at low evaluation counts
- Consistent gains for both DDPM and DDIM sampling regimes
- Improves quality without sacrificing tractable density estimation

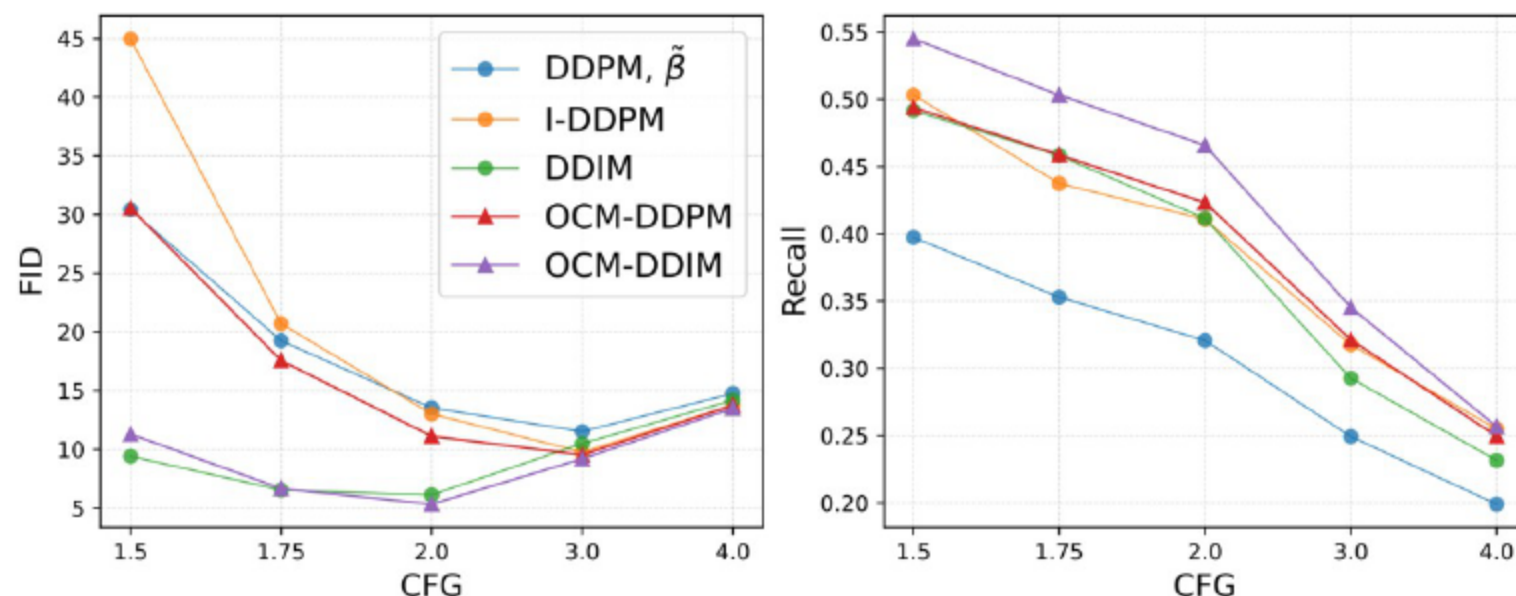
## 09 MoG: setup, evaluation, and results



- Two-dimensional mixtures with known scores enable explicit covariance error measurement
- OCM yields lower diagonal-covariance error and lower discrepancy with fewer denoising steps
- Full covariance outperforms diagonal on toys, motivating low-rank or block extensions
- Clear wins over prior diagonal learning approaches across step budgets

[Gretton et al., 2012]

## 10 DiT/CFG on ImageNet 256×256



- With 10 steps and classifier-free guidance near 2.0, OCM attains the best sample quality and higher recall
- Maintains tractable likelihood while improving diversity compared to deterministic-variance samplers
- Mean function from DiT is unchanged; covariance is learned via OCM
- Scales to large images with competitive efficiency

[Peebles & Xie, 2023; Sajjadi et al., 2018]

## 11 FID, steps, diversity

| METHOD         | CIFAR10                 | CELEBA 64x64            | LSUN BEDROOM            | IMAGENET 256x256        |
|----------------|-------------------------|-------------------------|-------------------------|-------------------------|
| DDPM           | 90 (6.12)               | > 200                   | 130 (6.06)              | 21 (5.89)               |
| DDIM           | 30 (5.85)               | > 100                   | BEST FID > 6            | 11 (5.58)               |
| IMPROVED DDPM  | 45 (5.96)               | MISSING MODEL           | <b>90</b> (6.02)        | 22 (6.08)               |
| ANALYTIC-DPM   | 25 (5.81)               | 55 (5.98)               | 100 (6.05)              | MISSING MODEL           |
| NPR-DPM        | 1.002×23 (5.76)         | 1.013×50 (6.04)         | 1.021× <b>90</b> (6.01) | MISSING MODEL           |
| SN-DPM         | 1.005×17 (5.81)         | 1.019×22 (5.96)         | 1.114×92 (6.02)         | MISSING MODEL           |
| OCM-DPM (OURS) | 1.003× <b>16</b> (5.83) | 1.015× <b>21</b> (5.94) | 1.112× <b>90</b> (6.04) | 1.007× <b>10</b> (5.33) |

- Fewest steps to reach an FID near six in most benchmarks after accounting for small head overhead
- Competitive or improved likelihood relative to recent moment-based methods at similar cost
- Better quality-diversity trade-off across CIFAR-10, CelebA, LSUN, and ImageNet
- Only method among strong baselines delivering both likelihood and sample quality consistently

[Heusel et al., 2017]



## 12 Covariance choices and applicability

Modeling Capability

| Covariance Type                                                | +#Passes | Intuition                     |
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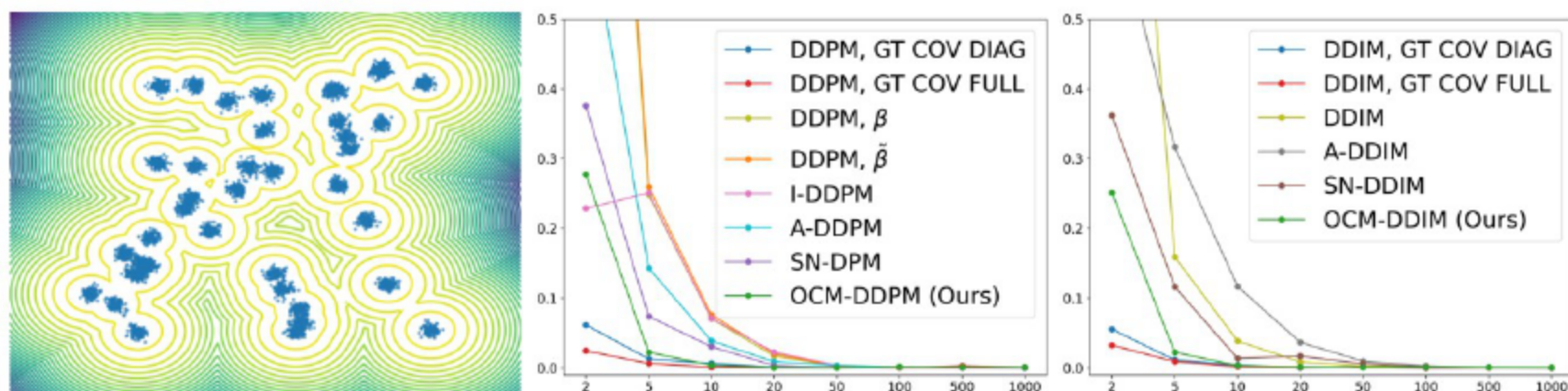
- OCM versus heuristic betas, analytic isotropic, variational lower bound learning, and noise-moment methods
- OCM applies cleanly to DDPM skip steps and to DDIM with zero-step targets, avoiding ill-defined cases
- Uses second-order score information and reduces error amplification near the noise-free limit [Zhang et al., 2024b; Bao et al., 2022a]
- Broadly compatible with pretrained diffusion backbones and schedules [Bao et al., 2022b]

## 13 Broader acceleration and distillation

| # timesteps           | FiD ↓        |              |              |             | NLL ↓       |             |             |             |
|-----------------------|--------------|--------------|--------------|-------------|-------------|-------------|-------------|-------------|
|                       | 5            | 10           | 15           | 20          | 5           | 10          | 15          | 20          |
| DDPM, $\tilde{\beta}$ | 58.28        | 34.76        | 24.02        | 19.00       | 203.29      | 74.95       | 44.94       | 32.20       |
| DDPM, $\beta$         | 254.07       | 205.31       | 149.67       | 109.81      | 7.33        | 6.51        | 6.06        | 5.77        |
| OCM-DDPM              | <b>38.88</b> | <b>21.60</b> | <b>13.35</b> | <b>9.75</b> | <b>6.82</b> | <b>4.98</b> | <b>4.62</b> | <b>4.43</b> |

- Complementary to fast continuous-time solvers and trainable forward processes such as bridge methods
- Distillation offers few-step generation but often lacks tractable likelihood **limiting downstream use**
- OCM preserves tractable density, aiding compression and other likelihood-critical tasks [Zhang et al., 2024b; Bao et al., 2022a]
- Can pair with expressive decoders or alternative conditionals for further gains [Wang et al., 2021; De Bortoli et al., 2021]

## 14 Beyond diagonal; applications



- Diagonal covariance limits fidelity; explore low-rank or block-diagonal structures for better trade-offs
- Applications: scientific modeling, likelihood-based compression, video generation, and inverse problems
- Higher-order moment matching could **accelerate** distillation and improve stability [Zhang et al., 2024a; Chung et al., 2022]
- Integrating improved covariance into general stochastic and deterministic solvers is promising [Vargas et al., 2023; Chen et al., 2024]

THANKS!